FACA: Force-Aware Contact Adaptation for Task-Parameterized Robotic Assembly

Xing Hu* **Xiangkang Zhao*** **Jinxin Liu** **Runzhe Zhan** **NaiJin Liu**
Xi'an Jiaotong University
xinghu@stu.xjtu.edu.cn
*Equal contribution.

Abstract:
Reliable axial assembly requires combining nominal task knowledge with feedback about execution-dependent contact conditions. We introduce FACA, a task-conditioned, contact-aware world model for parameterized robotic assembly. FACA encodes nominal task descriptors and recent action–response histories containing proprioception and wrist force/torque through separate pathways, then integrates them into a recurrent latent state. During simulation training, lateral-alignment regression and force-regime classification supervise the contact representation. At execution time, new measurements update the latent state, and task-conditioned latent dynamics propagate it for model predictive control without requiring future sensor measurements. Across ten assembly families evaluated at held-out sizes, FACA achieves a success rate of 76.1%, compared with 66.0% for task-conditioned TD-MPC2. On a real robot, optional residual-dynamics calibration achieves 82.0% success (41/50) with correction-assisted collection and 68.0% (34/50) with autonomous collection, using 740 target-domain transitions in each case. These results support combining nominal task context with recent interaction evidence for assembly planning and control.

Keywords: Reinforcement Learning, World Models, Force Sensing, Robotic Assembly

1 Introduction

Contact-rich axial assembly depends on both a static nominal task context and the contact condition that develops during execution. The task descriptor specifies mating type, geometry, clearance, and nominal axial extent, while execution-time effects such as grasp or fixture offsets and friction are reflected indirectly in motion and force/torque. A robot must therefore use the task descriptor as a prior and recent interaction evidence to update its control representation online. We study this setting in parameterized axial assembly.

The central challenge is that nominal task specifications do not uniquely determine the actual contact condition. Even when the mating type, geometry scale, clearance, and target insertion depth are known, grasp offset, fixture tolerance, residual misalignment, contact-mode transitions, friction variation, and jamming can still cause failures. These execution-time conditions are not specified by the task descriptor and are only indirectly reflected in robot motion and force/torque response. Prior force- and compliance-based insertion studies show that tight clearance and millimeter-scale pose uncertainty can cause assembly failure [1, 2, 3]. We use recent action–response history, including proprioception and wrist force/torque, to extract online evidence about the evolving contact condition.

We propose FACA, a task-conditioned, contact-aware world model for parameterized axial assembly. Architecturally, FACA routes the known nominal task descriptor p and newly measured exe-

10th Conference on Robot Learning (CoRL 2026), Austin TX, USA.

cution history through distinct conditioning pathways before latent-state update. The descriptor encodes the intended assembly condition, including mating type, geometry scale, clearance, and nominal axial extent. Recent action–response history, including proprioception and wrist force/torque, provides online contact evidence that is integrated with nominal task context into a contact-aware latent state z_t. This state conditions latent dynamics, reward prediction, value estimation, and action selection. The task pathway supplies a static nominal prior, while the history pathway supplies time-varying action–response evidence for prediction and planning.

We evaluate FACA on ten AutoMate [4]/SRSA [5]-derived axial assembly tasks with held-out size conditions. Nominal parameters are treated as a given task prior; the central experimental question is how execution history and contact-aware latent modeling improve control when the execution-time contact condition is not directly observed. Matched task-conditioned TD-MPC2 [6] variants isolate the additions of history, separate task/interaction conditioning pathways, and auxiliary supervision. A separate descriptor study in Appendix E.3 tests held-out nominal-parameter generalization. We also report contact-force statistics and protective stops; supplementary contact diagnostics are provided in Appendix D.3.

Our contributions are:

- **Separate task and interaction conditioning pathways.** FACA encodes nominal task descriptors and recent force–proprioceptive interaction history through distinct pathways and fuses them in a recurrent, task-aware state update. The static task context also explicitly conditions prediction and planning, while measurement-dependent contact context is carried through the latent state. This separation concerns the directly supplied inputs; it does not imply an identifiable decomposition of physical parameters.

- **Auxiliary supervision for contact-aware latent control.** We use lateral-alignment and force-regime supervision during simulation training to shape the contact representation built from recent action–response history. This contact evidence is integrated into a recurrent latent state for task-conditioned model predictive control.

- **Matched component evaluation and real-world validation.** We compare task-conditioned TD-MPC2 variants with matched action spaces and training protocols to measure the gains associated with execution history, separate task/interaction conditioning pathways, and auxiliary supervision, and validate the model on real hardware, including limited-data residual-dynamics calibration.

2 Related Work

2.1 Generalizable Robotic Assembly

Early learning-based assembly methods acquired contact-rich skills through guided policy search or used CAD models to guide policy learning [7, 8]. AutoMate extends this setting to diverse plug–socket geometries, learning specialist policies in simulation and combining policy distillation with reinforcement learning to obtain generalists [4]. SRSA addresses transfer to new assemblies by predicting the zero-shot performance of library skills, retrieving a suitable policy, and fine-tuning it on the target task [5]. These two works directly motivate our task setting and assembly assets.

Our focus is the representation used during execution of related axial assemblies. FACA learns a shared world model conditioned on known mating type, scale, clearance, and axial extent. A separate history pathway integrates the observed action–response evidence needed to account for contact conditions that the nominal descriptor cannot specify. This complements policy distillation and skill retrieval by examining how nominal task context and changing interaction evidence support predictive control.

2.2 Force and Tactile Feedback for Contact-Rich Assembly

Hybrid position/force and impedance control provide foundations for regulating motion and interaction jointly [9, 10, 11]. Learning-based insertion methods adapt compliance to contact conditions [3]. FORGE combines force-aware policy learning, dynamics randomization, and an allowable-force condition to handle pose uncertainty [1]. Chen et al. use compliant interactions to localize an uncertain target hole and exploit environmental constraints to refine insertion motions [12]. Both are closely related to our motivation that execution-time contact provides information beyond a nominal target specification.

Tactile-RL learns pose corrections between insertion attempts and studies how tactile representations affect transfer to unfamiliar geometries [13]. FoAR combines visual and force/torque features with a future-contact predictor and reactive action refinement [14]. ForceVLA fuses real-time force feedback with visual-language features during action decoding [15], while ForceVLA2 uses force-aware task concepts and interaction forces for hybrid force-position regulation [16]. These methods motivate integrating contact evidence into action generation. FACA uses that evidence to condition a predictive latent state for planning: it pairs recent executed actions with their subsequent state and wrist-wrench responses to update a latent control state. Lateral-alignment and force-regime targets supervise the contact pathway during simulation training; the auxiliary outputs are not used to score candidate actions at execution.

2.3 Contact-Aware World Models

PlaNet learns latent dynamics for planning from visual observations [17]. Dreamer and DreamerV3 learn control behaviors through latent imagination [18, 19], while DayDreamer applies world-model learning directly to physical robots [20]. TD-MPC combines short-horizon latent planning with a learned terminal value, and TD-MPC2 extends this decoder-free approach to scalable continuous control [21, 6]. FACA uses the TD-MPC2 planning backend; our matched task-conditioned variants examine the effects of interaction history, separate conditioning pathways, and contact supervision on its control representation.

The closest insertion studies already connect sensory feedback to world models. Palenicek et al. use DreamerV3 for tactile insertion in simulation and on a real robot [22]. Hansen et al. combine wrist-camera observations and proprioception in a shared insertion world model to generalize to unseen objects with unknown geometry [23]. OmniVTA predicts short-horizon visuo-tactile contact evolution and combines it with contact-aware fusion and reflexive feedback [24]. These works establish precedents for sensory world models in contact-rich control. FACA targets parameterized axial assembly with known nominal descriptors: static task context explicitly conditions prediction and planning, while measured force-proprioceptive history updates the recurrent state. Alignment and force-regime supervision are applied to the contact context before latent fusion. As in latent predictive control generally, future rollout proceeds from the current encoded state without future sensor measurements.

2.4 Sim-to-Real and Human-Assisted Adaptation

Dynamics randomization broadens the conditions encountered in simulation, and SimOpt adapts the randomization distribution using real-world rollouts [25, 26]. For contact-rich skills, Zhang et al. learn residual admittance parameters online from force measurements [27]. Residual reinforcement learning and SPARR instead learn action corrections on top of an existing controller or simulation-trained policy [28, 29]. Human correction also supports residual-policy learning in TRANSIC and real-world policy learning in HIL-SERL [30, 31].

Simulation Distillation is particularly close to our optional calibration: it adapts latent dynamics from real interaction while retaining simulation-trained representations and reward/value models for planning [32]. FACA implements adaptation as an additive latent-transition correction with the encoder, base dynamics, reward/value models, and policy prior frozen. Our calibration study com-

pares autonomous and correction-assisted transition collection under the same data budget. Human actions change the collected transitions, which train the residual dynamics adapter; this study therefore concerns how target-domain data supports predictive-model calibration.

3 Method

AutoMate [4] provides diverse assembly geometries and simulation environments for learning plug–socket assembly policies. Building on this setting, we consider parameterized axial assembly with known nominal task specifications and execution-dependent contact conditions. Our goal is to learn a shared task-conditioned, contact-aware world model that combines nominal task priors with recent interaction evidence for prediction and planning across task parameters.

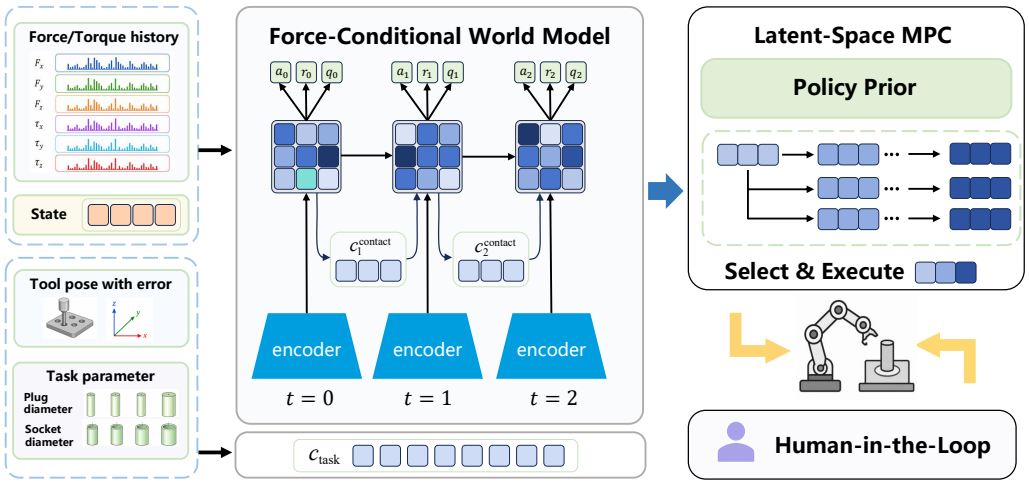


Figure 1: FACA architecture. Nominal task descriptors and measured action–response history update a contact-aware latent state for MPC and policy-prior action selection. Plug/socket follow held/fixed asset naming; diameter labels denote nominal geometry proxies. Human intervention supports optional correction-assisted data collection.

3.1 Nominal Task Parameters and Contact-Aware Latent State

The five-dimensional nominal task descriptor is

$$p = \left[t_{\text{type}},\ \log\left(\frac{D}{D_{\text{ref}}}\right),\ \frac{c}{c_{\text{ref}}},\ \frac{2c}{D},\ \frac{d}{d_{\text{ref}}} \right] \in \mathbb{R}^5, \tag{1}$$

where $t_{\text{type}} = 0$ denotes peg-in-hole and $t_{\text{type}} = 1$ denotes hole-onto-peg, D is the held-part transverse-width proxy, c is the configured radial-clearance descriptor, and d is the nominal axial extent of the held-part mesh, used as a geometric scale feature. Plug/socket denote held/fixed asset roles. Descriptor values remain fixed within an episode. The reference scales and task splits are given in Appendices A.2 and A.1, respectively.

At time t, the state input s_t comprises the TCP pose, linear and angular velocities, gripper width, and known nominal target position. Relative TCP position is computed against a nominal target reference, with reference-position error sampled once per episode and held fixed. Reference construction is specified in Appendix A.4. The wrist wrench w_t contains measured force and torque. The runtime observation is $o_t = [s_t, w_t]$, and the action a_t specifies a three-dimensional Cartesian displacement. We define the four-step execution history as

$$\tau_t^4 = \big((s_i, w_i, a_{i-1})\big)_{i=t-3}^{t}. \tag{2}$$

Each action is paired with its subsequent state and wrench response. This history provides contact evidence for the current latent-state update. Input specifications, wrench preprocessing, and history initialization are detailed in Appendices A.4 and C.2.

The nominal descriptor p specifies the intended assembly condition, while execution-time effects—such as fixture offset, grasp error, friction mismatch, sliding, jamming, and progress stagnation—are reflected indirectly in the observed motion and contact response. Rather than explicitly estimating these physical residuals, FACA updates a task-conditioned, contact-aware latent state z_t by integrating the current state, previous latent state and action, nominal task context, and contact evidence from recent action–response history. This recurrent state supports dynamics prediction and action selection and serves as a learned belief-like summary of execution-relevant information, but it is not a calibrated posterior over residual physical parameters. During simulation training, an auxiliary head predicts a separately defined alignment target; this does not assign a physical parameter interpretation to the latent coordinates.

3.2 Task-Conditioned Contact-Aware World Model

FACA encodes nominal task descriptors and recent force–proprioceptive interaction history through distinct pathways, and fuses them into a recurrent latent state for task-conditioned prediction and planning. This separation concerns the directly supplied inputs, rather than an identifiable decomposition of physical parameters.

The known task descriptor is encoded as a static task context:

$$c_{\text{task}} = T_\psi(p). \tag{3}$$

A feed-forward MLP H_κ summarizes the execution history as a feature h_t, which the contact encoder C_η uses to construct the contact context:

$$h_t = H_\kappa(\tau_t^4), \tag{4}$$

$$c_t^{\text{contact}} = C_\eta(s_t, h_t, z_{t-1}, a_{t-1}). \tag{5}$$

Temporal recurrence is provided by z_{t-1}, and wrist-wrench evidence enters through h_t. Network dimensions and hidden layers are given in Appendix B.1.

The current latent state is updated by fusing nominal task context and online contact evidence:

$$z_t = E_\phi(s_t, z_{t-1}, a_{t-1}, c_{\text{task}}, c_t^{\text{contact}}). \tag{6}$$

Given z_t, the latent-control model predicts:

$$\hat{z}_{t+1} = D_\phi(z_t, a_t, c_{\text{task}}), \tag{7}$$

$$\hat{\mathbf{r}}_t = R_\phi(z_t, a_t, c_{\text{task}}), \tag{8}$$

$$a_t^\pi \sim \pi_\theta(\cdot \mid z_t, c_{\text{task}}), \tag{9}$$

$$\hat{Q}_t = Q_\phi(z_t, a_t, c_{\text{task}}). \tag{10}$$

The policy prior is a tanh-squashed Gaussian. Reward logits are decoded to a scalar for planning, and Q_ϕ denotes the decoded scalar action value; the output representation is specified in Appendix B.1.

The two conditioning pathways intentionally play asymmetric roles. The task context is static within an episode and known throughout the planning horizon. It enters the measured-state update and remains an explicit condition for the latent dynamics, reward, value, and policy-prior modules. This lets the recurrent update interpret action–response evidence relative to the nominal task geometry. In contrast, the contact context depends on measurements available at the current real control step. It is fused into z_t, which carries the interaction evidence forward during imagined rollout without requiring future measured wrench or future contact-context estimates.

To supervise the contact context that enters the measured-state update, we use two auxiliary heads:

$$\hat{e}_t = G_e(c_t^{\text{contact}}), \tag{11}$$

$$\boldsymbol{\ell}_t^m = G_m(c_t^{\text{contact}}), \tag{12}$$

where $\hat{e}_t \in \mathbb{R}^2$ predicts the held part's lateral displacement in the ground-truth fixed-part mating frame (metres), and $\ell_t^m \in \mathbb{R}^3$ contains logits for free, nominal-contact, and high-force regimes. Appendix B.2 defines the labels and their force channel. These heads directly supervise the measured contact context before latent fusion during training; their architecture is specified in Appendix B.2. Because this context enters E_ϕ, their losses also shape a representation used by the controller. The context is not a force-only feature: C_η also receives s_t, z_{t-1}, and a_{t-1}. The labels are not inputs at test time. The auxiliary predictions are training outputs and are not used to score candidate actions in the base controller.

The TD-MPC2 planning backend [6] scores a candidate action sequence $\mathbf{a}$ using

$$\mathcal{S}_t(\mathbf{a}) = \sum_{h=0}^{H_{\mathrm{mpc}}-1} \gamma^h \hat{r}(\hat{z}_{t+h|t}, a_{t+h|t}, c_{\mathrm{task}}) + \gamma^{H_{\mathrm{mpc}}} V_{\mathrm{term}}(\hat{z}_{t+H_{\mathrm{mpc}}|t}, c_{\mathrm{task}}). \tag{13}$$

The rollout starts at $\hat{z}_{t|t} = z_t$. Here $\hat{r}$ is decoded from the reward logits. The terminal value averages two randomly selected online critics at an action sampled from the policy prior. Elite-weighted refinement and final action selection are specified in Appendix C.4.

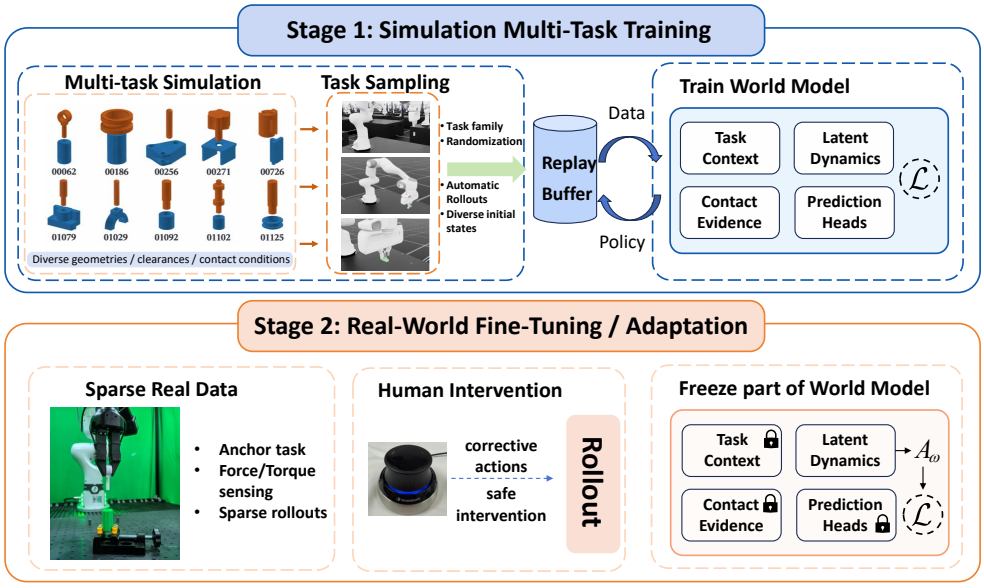


Figure 2: FACA training and calibration. Simulation learns the base world model. Ten real-world episodes on one anchor task provide 740 transitions for autonomous or correction-assisted calibration. Each calibrated adapter is shared across ten evaluation tasks.

3.3 Simulation Training

We train a shared FACA model across the task-parameter conditions in Appendix A.1, using a world-model update and a separate policy-prior update. Replay stores actions, state and wrench observations, rewards, and task descriptors. The simulator additionally provides lateral-alignment targets e_t and force-regime labels m_t for contact-context supervision.

For world-model learning, the encoder computes measured latent states from replay action–response histories. Starting from the first measured encoding in the loss window, D_ϕ rolls out imagined states $\hat{z}_t$ under recorded actions. Each predicted successor is matched to the detached measured encoding:

$$\ell_{\mathrm{dyn},t} = \frac{1}{d_z} \left\| \mathrm{sg}\left(z_{t+1}^{\mathrm{enc}}\right) - D_\phi(\hat{z}_t, a_t, c_{\mathrm{task}}) \right\|_2^2. \tag{14}$$

Here sg denotes stop-gradient and d_z is the latent dimension. Batch and temporal aggregation yield $\mathcal{L}_{\text{dyn}}$; replay reconstruction and aggregation details are given in Appendix C.2.

The model update follows the TD-MPC2 dynamics, reward, and Q objectives [6]. The two auxiliary heads additionally train the measured contact context for lateral-alignment regression and force-regime classification, with per-sample losses

$$\ell_{\text{align},t} = \tfrac{1}{2}\|e_t - \hat{e}_t\|_2^2, \qquad \ell_{\text{contact},t} = \text{CE}(m_t, \boldsymbol{\ell}_t^m). \tag{15}$$

Averaging these auxiliary terms over valid training samples gives $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{\text{contact}}$. The complete model objective is

$$\mathcal{L}_{\text{model}} = \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_r\mathcal{L}_r + \lambda_Q\mathcal{L}_Q + \alpha\mathcal{L}_{\text{align}} + \beta\mathcal{L}_{\text{contact}}. \tag{16}$$

The policy prior is updated separately to propose candidate actions for planning, using

$$\mathcal{L}_{\text{policy}} = \mathcal{L}_{\text{BC}} + \mathcal{L}_{\text{MaxQ}} + \mathcal{L}_{\text{entropy}}. \tag{17}$$

This combines imitation of replay actions with value-seeking and entropy regularization. The update uses detached imagined states and frozen Q networks, retaining their action gradients. Appendix C.3 gives the inherited reward/Q losses, coefficients, and update details.

Environment rewards are specified in Appendix C.1.

3.4 Optional Target-Domain Calibration

For real-world deployment, we optionally fit a lightweight residual correction to the frozen latent dynamics using a small set of target-domain transitions. Given the current latent state, executed action, and task context, the adapted one-step prediction is

$$\hat{z}_{t+1}^{\text{adapt}} = D_\phi(z_t, a_t^{\text{exec}}, c_{\text{task}}) + A_\omega^{\text{eff}}(\upsilon_t), \tag{18}$$

where υ_t collects the configured inputs available at planning time. Only the residual correction is fitted to the frozen encoder representation of the observed successor state; the task encoder, latent-state encoder, base dynamics, reward/value models, and policy-prior parameters remain fixed. Appendix F.1 details the objective and its application during planning.

4 Experiments

We evaluate how execution history, separate conditioning pathways and auxiliary supervision improve task-conditioned TD-MPC2 on axial assembly. Task-conditioned variants share the descriptor representation defined above. Appendix E.3 separately evaluates descriptor generalization to held-out nominal parameters.

4.1 Experimental Setup

Tasks and protocol. Experiments use ten AutoMate/SRSA-derived axial assembly tasks in Isaac Sim, with the same assembly families used for training and testing at disjoint size conditions. Each assembly trains on one of two five-scale sets (A or B) and is tested on the complementary set. This yields 50 training and 50 held-out test conditions, including 40 interpolation and 10 extrapolation conditions. All numerical baselines are evaluated on this test split in our environment. Task assets, scale sets, per-assembly assignments, and initial-pose distributions are given in Appendix A.

The matched comparison shares tasks, initialization, rewards, interaction budgets, seeds, controller and evaluation protocol. Actions are 3D Cartesian translations with controller-maintained orientation.

Table 1: Simulation baselines and component ablations on held-out sizes. M0–M3 share nominal task conditioning, and M1–M3 use the same history window. SR follows the terminal criterion in Appendix D. FP averages normalized terminal insertion depth over all trials, clipped to $[0, 1]$ per trial. $\overline{F}_{\text{peak},\, J=0}$ is mean episode-peak force over trials without a detected near-jam event. Complete outcomes are in Table 7.

Method	Backend	Interaction history	Separate contact encoder	Aux. sup.	SR (%)↑	FP↑	$\overline{F}_{\text{peak},\, J=0}$ (N)↓
AutoMate-style PPO	PPO	–	–	–	54.1	0.63	20.9
Task-PPO	PPO	–	–	–	53.6	0.65	20.1
Task-TD-MPC2 (M0)	TD-MPC2	–	–	–	66.0	0.73	19.6
+ recurrent history (M1)	TD-MPC2	✓	–	–	69.0	0.77	16.4
+ contact encoder (M2)	TD-MPC2	✓	✓	–	71.8	0.79	13.5
FACA (M3)	TD-MPC2	✓	✓	✓	**76.1**	0.84	**12.9**

Metrics. SR is task-averaged success rate: the final three control samples must reach at least 85% target insertion depth and satisfy the clearance-dependent lateral tolerance. Geometric success does not terminate an episode. FP averages terminal depth normalized by target depth and clipped to $[0, 1]$ over all trials. MF, denoted by $\overline{F}_{\text{peak},J=0}$ (N), averages episode-peak force over trials without a detected near-jam event. Near-jam requires four consecutive plug–socket contact-force samples ≥ 15 N and first-to-last insertion-depth increase < 0.1 mm (0.20 s at 15 Hz). A trial has $J = 1$ if any event occurs; otherwise $J = 0$. MF does not summarize all-trial force exposure. Definitions and JR results are in Appendices D.3 and E.

4.2 Simulation Results and Component Ablations

Table 1 compares PPO baselines and TD-MPC2 variants on held-out sizes. PPO baselines differ in nominal task conditioning. Starting from task-conditioned M0, successive variants add recurrent state/action/wrench history (M1), a contact encoder (M2), and alignment/force-regime supervision (M3/FACA).

FACA achieves the highest reported SR in Table 1, reaching 76.1% compared with 66.0% for M0, a gain of 10.1 percentage points. The two PPO baselines reach 54.1% and 53.6%, respectively.

Within the TD-MPC2 sequence, M1 reaches 69.0% SR after adding interaction history and recurrence, a 3.0-point increase over M0. M2 introduces a separate contact encoder before latent fusion, retaining the same history window and reaching 71.8% (+2.8 points). Enabling the two auxiliary losses in M3 raises SR to 76.1% (+4.3 points). These increments quantify the gains associated with the component groups added at each stage.

FACA also increases FP from M0's 0.73 to 0.84, indicating greater mean terminal axial insertion progress across evaluated trials. For trials without a detected near-jam event, mean episode-peak force is 12.9 N for FACA and 19.6 N for M0. These force values describe each method's $J = 0$ subset; full contact-event results are provided in Table 7.

In the independent descriptor study on two assembly families with a force-free backbone, the full descriptor improves held-out average SR from 39.3% without an explicit descriptor branch to 54.3%. The settings and complete results are given in Appendix E.3.

A separate policy-transfer comparison gives 68.3% SR for SRSA-style transfer and 76.1% for FACA, a gap of 7.8 percentage points. This compares complete methods. Supplementary outcomes are in Table 7; Figure 7 in Appendix E.1 shows the overall and per-task policy comparisons.

4.3 Real-World Target-Domain Calibration

We further evaluate FACA on an FR3 robot with a wrist force/torque sensor (Figure 3). Relative TCP position uses an operator-taught assembly-start reference, with simulation-matched reference error sampled once per trial and held fixed. We compare direct transfer (Sim-only), the fine-tuning

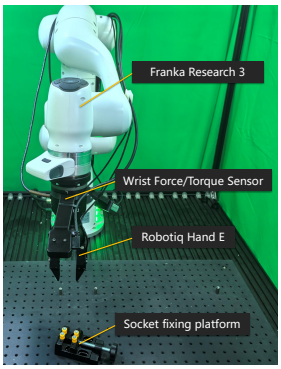


Figure 3: FR3 setup.

Table 2: Real-robot calibration (50 trials/method). FP: mean terminal depth fraction over all trials, clipped to $[0, 1]$ per trial; PS: protective-stop rate. $\overline{F}_{\text{peak},\, J=0}$: mean episode-peak force for $J = 0$ (no detected near-jam event). Both event detection and peak-force calculation use the tared wrist-force signal. Evaluation details are in Appendix F.2.

Method	Success ↑ ($n/50$, %)	FP↑	PS ↓ (%)	$\overline{F}_{\text{peak},\, J=0}$ (N) ↓
Sim-only	26/50 (52.0%)	0.61	36.0	42.5
Full-FT	30/50 (60.0%)	0.69	30.0	35.3
Auto-adapt	34/50 (68.0%)	0.74	24.0	34.8
HIL-adapt	**41/50 (82.0%)**	0.86	**10.0**	**27.6**

control (Full-FT), and optional residual-dynamics calibration using autonomous (Auto-adapt) or correction-assisted (HIL-adapt) transition collection. Full-FT updates only the encoder and base dynamics. Auto-adapt and HIL-adapt each collect ten episodes on a single anchor task, yielding 740 transitions in total, and use the same calibration objective. Each calibrated adapter is shared across all ten evaluation tasks. HIL-adapt differs only in collection: up to 111 operator actions replace autonomous proposals near predefined difficult contact states. Full-FT and Auto-adapt use 2,000 gradient updates on the same autonomous data. Each method is evaluated in 50 trials across 10 tasks and 10 sessions. Hardware success requires completed insertion to remain stable for 1 s; Appendix F.1 gives the detailed protocol.

Table 2 shows 26/50 (52.0%) successes for direct transfer, 30/50 (60.0%) for Full-FT, 34/50 (68.0%) for Auto-adapt, and 41/50 (82.0%) for HIL-adapt. Protective-stop rates fall from 36.0% to 30.0%, 24.0%, and 10.0%, respectively. Auto-adapt exceeds Full-FT using the same autonomously collected dataset and number of gradient updates. Correction-assisted collection improves the calibrated model from 34/50 to 41/50 successes under the same transition budget and objective. This result supports targeted transition acquisition near difficult contact states in this setup; the correction and failure protocols are detailed in Appendix F.

5 Limitations

FACA currently controls three-dimensional Cartesian translations, with orientation maintained by the low-level controller; rotational actions remain unexplored. The ten evaluated AutoMate-derived assemblies cover a limited portion of the full collection. Future work will study joint translation and rotation, more complex mating geometries, non-axial and multi-stage assembly, and broader real-robot task coverage. The method also assumes known nominal task descriptors, motivating perception-based parameter estimation.

6 Conclusion

FACA combines nominal task context and action–response history through separate pathways in a recurrent contact-aware world model for axial assembly. Alignment and force-regime supervision shape the contact representation. Measurements refresh the latent state at execution; imagined rollouts propagate it without future wrench measurements.

Matched TD-MPC2 variants improve simulation success from 66.0% to 76.1%; the descriptor study supports held-out parameter generalization. Hardware calibration achieves 41/50 successes (82.0%) with correction-assisted collection versus 34/50 (68.0%) with autonomous collection, using the same objective and 740 anchor-task transitions per method. Each adapter serves ten tasks. These results support contact-aware planning and targeted dynamics calibration.

Acknowledgments

We thank Yumin Chen for assistance with the robotic hardware platform, and Wenqian Xu for helpful discussions and constructive feedback during the development of this work.

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Supplementary Material

This supplement provides task and input specifications (A), network and supervision details (B), training and planning protocols (C), evaluation definitions (D), additional simulation results (E), and real-robot calibration and evaluation (F).

A Simulation Tasks and Inputs

A.1 Assembly Assets and Train–Test Size Splits

The ten assemblies in Table 3 include peg-in-hole and hole-onto-peg tasks, encoded by $t_{\text{type}} = 0$ and 1, respectively. Asset names *plug* and *socket* denote the held and fixed parts; their physical mating roles depend on task type. The held-part transverse width is $D = sD_0$, where D_0 is its maximum nominal XY bounding-box width. The clearance descriptor c is an asset configuration parameter. The nominal axial extent d is computed from the held mesh's bounding box along its assembly-aligned Z axis and used as a geometric scale feature. The template-defined mating-size proxy $D + 2c$ is not an independently measured aperture.

The two scale sets are $A = \{0.875, 0.925, 0.975, 1.025, 1.075\}$ and $B = \{0.900, 0.950, 1.000, 1.050, 1.125\}$. Each assembly trains on the set listed in Table 3; the complementary set defines its held-out test conditions. This gives 50 training and 50 held-out conditions, including 40 interpolation and 10 extrapolation conditions relative to each assembly's training range. The separate two-family descriptor study uses the splits in Appendix E.3.

Table 3: Base-assembly parameters at $s = 1$ and training scale assignments in the ten-task simulation study. D_0 is the nominal held-part width proxy; c and d are the configured radial-clearance descriptor and nominal held-part axial extent, respectively. Lengths are in mm.

Assembly ID	Width proxy D_0	Clearance descriptor c	Axial extent d	Train set
00062	23.090	0.1407	56.490	A
00186	27.576	1.6418	13.214	B
00256	6.678	0.3852	34.880	A
00271	32.650	0.0770	41.828	B
00726	22.785	0.2950	31.898	A
01079	7.878	0.4984	35.282	B
01029	8.390	0.4759	55.931	A
01092	9.516	0.4430	37.133	B
01102	12.792	0.0551	44.648	A
01125	6.838	0.5663	28.616	B

The clearance descriptor is a nominal configuration parameter, rather than a measured minimum gap along the full insertion path or an achieved positioning accuracy. Each assembly trains on the listed scale set and is evaluated on the complementary set.

Figure 4 illustrates transverse scaling while retaining axial extent.

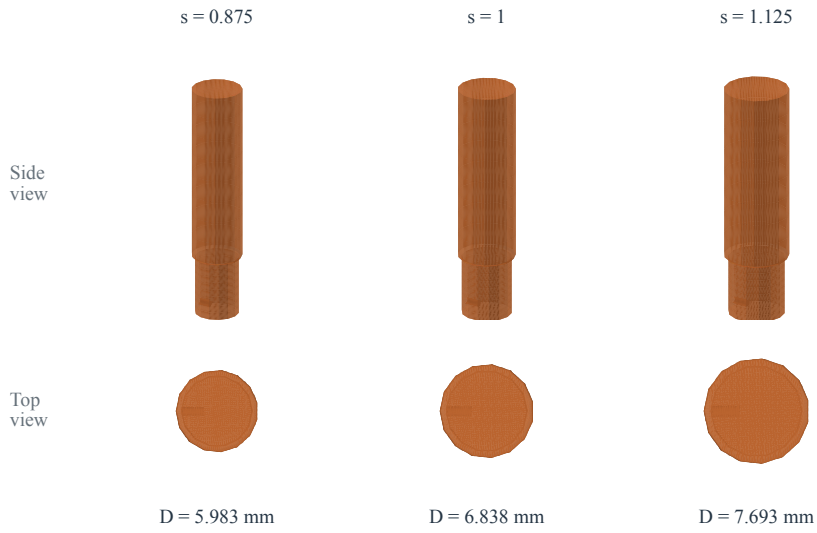
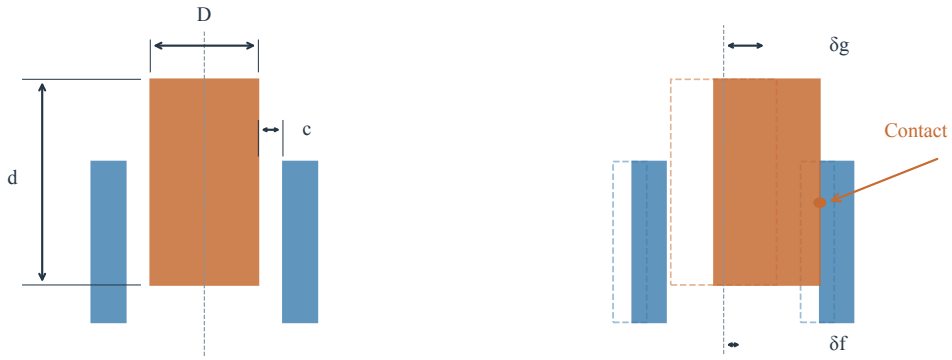


Figure 4: Transverse scaling of assembly 01125 at $s = 0.875$, 1.0, and 1.125 (S0, S5, and S9). Camera settings and display scales are fixed across scale variants within each row. D denotes the held-part XY width proxy; axial extent is unchanged. Training membership follows Table 3.

A.2 Nominal Descriptor and Execution Offsets

The five-dimensional descriptor in Eq. (1) encodes mating type, transverse width D, configured radial clearance c, and nominal axial extent d. All lengths are in millimetres. Fixed reference scales are $D_{\text{ref}} = 7.986$, $c_{\text{ref}} = 3.887$, and $d_{\text{ref}} = 28.616$; the latter two are the reference radius and axial extent of assembly 01125. These scales remain unchanged during training and evaluation. Type and continuous features are encoded separately before fusion (Appendix B.1). Figure 5 illustrates how grasp and fixture offsets change placement and contact relative to nominal geometry.



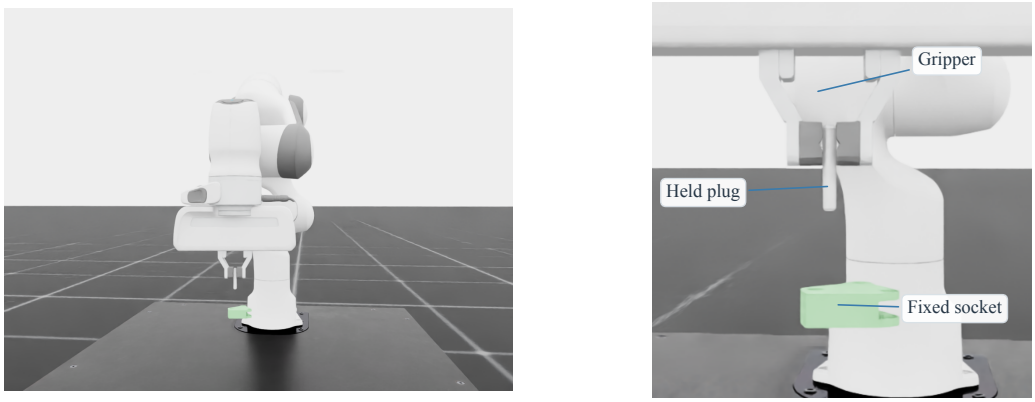
(a) Nominal geometry.

(b) Execution offsets.

Figure 5: Nominal geometry and execution offsets. Width D, configured clearance c, and nominal axial extent d describe the asset; grasp offset δ_g and fixture offset δ_f alter relative placement. Dashed outlines indicate nominal placement. Clearances and offsets are exaggerated for visibility and do not specify measured dimensions or an offset distribution. The geometric extent d is distinct from the target insertion depth h_{target}.

A.3 Simulation Setup and Initial Conditions

The gripper holds the moving mating part, while its counterpart remains fixed: the peg moves into a fixed hole in peg-in-hole tasks, and the hole moves onto a fixed peg in hole-onto-peg tasks. Supervision and evaluation use the ground-truth mating frame $\mathcal{F}$ attached to the fixed counterpart. The held-part position in this frame is $p_t^{\text{rel}} = R_{\mathcal{F}}^{\top}(p_{\text{held},t}^{W} - p_{\text{fixed}}^{W})$, where $R_{\mathcal{F}}$ maps frame coordinates to world coordinates, denoted by superscript W. Positive Z points toward the pre-insertion approach side, so the axial coordinate decreases during insertion. The policy controls three Cartesian translations $(\Delta x, \Delta y, \Delta z)$ at 15 Hz, with orientation maintained by the controller. All methods share the same reset distribution: X and Y offsets are sampled independently with uniform magnitudes from 1 to 9 mm and equiprobable signs; the Z interval is $[1, 2]$ mm and the yaw interval is $[-5^{\circ}, 5^{\circ}]$. Sampling-based curriculum (SBC) updates are disabled. Target-reference perturbations are applied separately from these initial-pose offsets, as described below. Figure 6 shows the apparatus at reset.



(a) Simulation overview.

(b) Gripper-normal view.

Figure 6: Simulation apparatus for assembly 00256 at reset. The close-up labels the gripper, held part (plug), and fixed counterpart (socket). Its optical axis follows the normal of the plane spanned by finger separation and TCP approach directions.

A.4 Observation and Wrench Preprocessing

The 17-dimensional state s_t combines 14 TCP/proprioceptive components (position, orientation, linear and angular velocity, and gripper width) with three known nominal target-position coordinates. The relative-position input uses a nominal target reference with an additive position error sampled once per episode and held fixed throughout that episode. The current TCP position and target reference use the same coordinate basis when their difference is computed. The six-dimensional wrench $w_t = (F_x, F_y, F_z, \tau_x, \tau_y, \tau_z)$ gives the runtime observation $o_t = [s_t, w_t] \in \mathbb{R}^{23}$; the translation action has three dimensions. The held-mesh position p_t^{rel} used for simulator labels and evaluation is distinct from the TCP pose in the runtime observation. Auxiliary alignment labels use the ground-truth fixed-part transform in Appendix A.3, rather than the perturbed runtime target reference.

In simulation, the wrench is a tared TCP joint-reaction measurement synchronized with the current state. Force and torque are normalized by $50\,\text{N}$ and $5\,\text{N}\,\text{m}$, respectively. The observation wrapper applies no temporal filter or clipping before the history buffer. This wrench channel is separate from the plug–socket pair force used for auxiliary labels and contact diagnostics. History alignment and initialization are specified in Appendix C.2.

B Network Architecture and Auxiliary Supervision

B.1 Network Configuration

Table 4 specifies FACA (M3), using the interfaces in Section 3.2. Brackets denote concatenation and vec flattening. Inputs s_t, w_t, and a_t have 17, 6, and 3 components (Appendix A.4). Feature widths 64 and 384 are architecture settings; input widths follow from concatenation.

Table 4: FACA network inputs, dimensions, and hidden widths. Hidden blocks use Linear–LayerNorm–Mish unless specified otherwise.

Module	Input	Input dim.	Hidden widths	Output dim.
Task T_ψ	p	5	32, 32; 64, 64	64
History H_κ	$\mathrm{vec}(\tau_t^4)$	104	128, 128	64
Contact C_η	$[s_t, h_t, z_{t-1}, a_{t-1}]$	468	128	64
State E_ϕ	$[s_t, z_{t-1}, a_{t-1}, c_{\text{task}}, c_t^{\text{contact}}]$	532	128	384
Dynamics D_ϕ	$[z, a, c_{\text{task}}]$	451	256, 256	384
Reward R_ϕ	$[z, a, c_{\text{task}}]$	451	256, 256	101
Q Q_ϕ	$[z, a, c_{\text{task}}]$	451	256, 256	3×101
Prior π_θ	$[z, c_{\text{task}}]$	448	256, 256	6
Alignment G_e	c_t^{contact}	64	128	2
Force class G_m	c_t^{contact}	64	128	3

Task encoding. Descriptor p contains mating type and four geometric features (Eq. (1)). Type selects an eight-dimensional embedding, while the continuous features follow a $4 \to 32 \to 32$ branch. Their concatenation gives $8 + 32 = 40$ inputs to a $40 \to 64 \to 64$ fusion branch, producing c_{task}.

History and measured-state encoding. Four chronologically ordered tuples (s_i, w_i, a_{i-1}) give $4(17 + 6 + 3) = 104$ history inputs, mapped to $h_t \in \mathbb{R}^{64}$ by a feed-forward MLP. The contact encoder concatenates current state, history feature, previous latent, and previous action: $17 + 64 + 384 + 3 = 468$. State fusion adds both contexts to state, previous latent, and previous action: $17 + 384 + 3 + 64 + 64 = 532$. History and contact outputs are linear; recurrence enters through z_{t-1}, and wrench enters through h_t.

Prediction and control heads. Dynamics, reward, and Q use $384 + 3 + 64 = 451$ inputs; the prior uses $384 + 64 = 448$. State and dynamics outputs use SimNorm in groups of eight. Reward and each Q critic output 101 logits. The prior outputs three means and three log standard deviations. Auxiliary heads read the measured contact context to predict lateral alignment and force regime (Appendix B.2).

Trainable parameter counts are 1,378,256 for M0, 1,473,168 for M1, and 1,656,661 for M2/M3. These count parameters with `requires_grad=True`, including M2 auxiliary heads with zero loss weights.

Reward and Q use 101 bins on the symlog support $[-10, 10]$, spaced by 0.2. Targets are symlog-transformed, clipped to the support, and linearly interpolated between adjacent bins to form two-hot labels. Decoding applies symexp to the softmax-weighted bin centre. Here $\mathrm{symlog}(x)$ is $\mathrm{sign}(x)\log(1 + |x|)$ and symexp is its inverse. The clipping affects distributional training labels, not environment reward. The prior uses a tanh-squashed Gaussian with log standard deviation in $[-10, 2]$.

B.2 Auxiliary Labels

The alignment target is the held part's ground-truth lateral displacement in the fixed-part mating frame $\mathcal{F}$. Using the relative position p_t^{rel} defined in Appendix A.3 and a nominal lateral target of

zero,

$$e_t = [p_t^{\mathrm{rel}}]_{xy}. \tag{19}$$

It is measured in metres without additional scaling. The alignment loss averages squared error over two coordinates and valid post-burn-in samples. Force-regime supervision uses the plug–socket pair-force norm after a causal three-sample median: free below 1 N, nominal contact for force magnitudes in $[1, 10)$ N, and high force at or above 10 N. At startup only available samples are used; two samples use the lower median. The class loss is cross entropy on logits. High force is not a mechanical-jam label.

Both auxiliary heads read the measured c_t^{contact} before latent fusion, with no detach, so their gradients propagate through C_η. Losses use valid post-burn-in samples with the recorded temporal alignment. These simulator labels are used only for training, and are not runtime policy inputs or MPC scoring terms.

C Training and Planning

C.1 Environment Reward

The matched simulation runs use the AutoMate reward, comprising signed distance field (SDF) alignment, reference-trajectory imitation, and training-success terms with equal relative weights. An SDF encodes signed distance to a mesh surface. The alignment reward queries the held-part mesh at its target pose using surface samples from the currently posed held part, providing a dense geometric alignment signal. The imitation term matches ten-position TCP histories to reversed-disassembly reference trajectories. These procedures follow the Isaac Lab AutoMate assembly environment [4]. The training-success term is issued after three valid samples, includes the axial bound $0 < u_z < h_{\mathrm{target}}$, and continues while the conditions hold. Training success is distinct from evaluation success. Here $u_z = [p_t^{\mathrm{rel}}]_z$ is the held part's axial coordinate in the fixed-part mating frame, and h_{target} is the runtime target insertion depth. The axial target is denoted separately from transverse width D and the nominal mesh axial extent d; d is a descriptor feature, while h_{target} specifies the insertion goal. No additional wrist-force penalty or final reward clipping is used. Reward geometry and reference trajectories are not policy observations.

C.2 Replay Reconstruction and Dynamics Targets

For M1–M3, training samples eight observations and seven aligned actions from one episode. The first four observations form a gradient-free burn-in; the remaining four encodings supervise three transitions. At the fragment's left boundary, previous latent and action are zero, and unavailable history is zero-padded, even for a fragment sampled inside an episode. No stored collection latent or earlier episode prefix is used. The burn-in endpoint is detached; subsequent measured encoding is recurrent without detaching the recurrent state at each step. Dynamics targets are detached outputs of the current encoder, not of a separate target encoder. M0 instead samples four observations for three transitions, without history or burn-in; it encodes each observation without recurrent state.

Let b be the first loss-window time, $\bar{z}$ the measured encoding, and $\hat{z}$ the imagined state. The rollout and mean-squared consistency loss are

$$\begin{aligned} \hat{z}_{b|b} &= \bar{z}_b, \\ \hat{z}_{b+k+1|b} &= D_\phi(\hat{z}_{b+k|b}, a_{b+k}, c_{\mathrm{task}}), \\ \mathcal{L}_{\mathrm{dyn}} &= \sum_{k=0}^{H-1} \omega_k \mathbb{E}_{\mathcal{B}} \left[\frac{\| \hat{z}_{b+k+1|b} - \mathrm{sg}(\bar{z}_{b+k+1}) \|_2^2}{d_z} \right]. \end{aligned} \tag{20}$$

Here $H = 3$ is the training rollout horizon in transitions, and $d_z = 384$ is the latent-state dimension. The temporal weights are $\omega_k = \lambda_{\mathrm{temp}}^k / \sum_{j=0}^{H} \lambda_{\mathrm{temp}}^j$ with $\lambda_{\mathrm{temp}} = 0.5$. The dynamics weights are $8/15, 4/15, 2/15$; their sum is $14/15$, since the denominator is shared with losses using $H + 1$

states. Later measurements supply targets but never replace imagined states in this rollout. This is the batch-and-time aggregation of Eq. (14).

Deployment resets recurrent state only at episode start; unavailable history is left-padded with zeros. Four-step burn-in is a finite-prefix approximation and need not reproduce the deployment state obtained from the full episode history.

C.3 Objectives and Update Order

The model objective in Eq. (16) combines dynamics, reward, Q, and auxiliary losses. Its M3 coefficients are $(\lambda_{\mathrm{dyn}}, \lambda_r, \lambda_Q, \alpha, \beta) = (20, 0.1, 0.1, 6.5, 0.072)$. M2 sets both auxiliary coefficients to zero. Dynamics and reward/Q use imagined states. Both M3 auxiliary heads read the measured contact context c_t^{contact}, before latent fusion. Table 5 specifies the gradient boundaries.

Table 5: Loss inputs and gradient boundaries for the contact-context-head architecture.

Loss	Input	Gradient boundary
Dynamics	Imagined successor; measured target	Target detached; burn-in without gradients
Reward / Q	Imagined state and replay action	TD target without gradients
Alignment / class	Measured contact context	Bypasses same-step fusion output; earlier fusion reachable through recurrence; no imagined-dynamics path
Policy prior	Detached imagined state	Frozen Q retains action derivative; prior parameters only

Following TD-MPC2 [6], reward and Q use two-hot cross entropy. Let $\hat{\boldsymbol{r}}_k$ and $\hat{\boldsymbol{q}}_{j,k}$ denote reward and online-Q logits evaluated at the imagined rollout state and replay action $(\hat{z}_k, a_k, c_{\mathrm{task}})$. With τ mapping a scalar to two-hot labels on symlog bins, the per-transition losses are

$$\begin{aligned} \ell_{r,k} &= \mathrm{CE}(\tau(r_k), \hat{\boldsymbol{r}}_k)\,, \\ \ell_{Q,k} &= \frac{1}{N_Q} \sum_{j=1}^{N_Q} \mathrm{CE}\big(\tau(\mathrm{sg}(y_k)), \hat{\boldsymbol{q}}_{j,k}\big)\,. \end{aligned} \tag{21}$$

Here $N_Q = 3$. The scalar TD target is computed without gradients, with the next action sampled from the policy prior at the detached measured successor encoding:

$$y_k = r_k + \gamma \min_{j \in \mathcal{I}_2} Q_j^{\mathrm{targ}}(\mathrm{sg}(\bar{z}_{k+1}), \tilde{a}_{k+1}, c_{\mathrm{task}}), \quad \tilde{a}_{k+1} \sim \pi_\theta. \tag{22}$$

In the matched runs $\gamma = 0.95$ and $\mathcal{I}_2$ samples two of three target critics. The reward and Q losses in Eq. (21) use soft cross entropy against two-hot labels, averaged over the batch and summed with temporal weights, without an additional division by the rollout horizon; Q loss additionally averages all three online critics. The training entry point disallows non-timeout terminations. The 74-step timeouts retain the final observation and bootstrap without a termination mask.

The separate policy objective in Eq. (17) uses sampled tanh-Gaussian actions. Its cloning term is

$$\mathrm{BC}(\tilde{a}_k, a_k) = \frac{\sum_i m_i (\tilde{a}_{k,i} - a_{k,i})^2}{\sum_i m_i}. \tag{23}$$

All three action dimensions are valid. BC has coefficient one and uses the first H states; normalized Max-Q and entropy use $H+1$ states with weights $(8, 4, 2, 1)/15$. Max-Q averages a shared random pair of online critics. Its scale S starts at one and is updated with rate 0.01 toward the first-state Q samples' 95th–5th percentile difference, floored at one; Q is divided by S without mean centering.

The entropy coefficient is 10^{-4}. With Gaussian log probability ℓ_G and tanh-corrected log probability ℓ_{sq}, the implemented scaled estimate is $-\ell_{\mathrm{sq}}\, 3\ell_G/(\ell_{\mathrm{sq}} + 10^{-8})$. It is not three times the squashed-policy entropy.

Updates proceed as model Adam, target-Q Polyak averaging, then policy Adam. Policy optimization reuses detached pre-update latents with updated online Q. Optimizer and sampling settings are given below.

C.4 Training and Planning Configuration

Planning follows the TD-MPC2 MPPI procedure [6]. At every real control step the newest measurements refresh the latent state; the measured latent and executed action are cached, not the last imagined latent. Planning terminal value averages two randomly selected online critics, whereas TD backup uses the minimum of two target critics. Evaluation removes extra final Gaussian exploration noise but retains policy and elite sampling. Proposal initialization uses the mean and standard deviation of policy-prior trajectories. The shifted-proposal mixture weight is zero in these runs. Of the 512 candidates, 24 are policy-prior trajectories and the remaining 488 are Gaussian proposals.

Table 6: Simulation training and MPC hyperparameters for M0–M3.

Training		MPC planning	
Parameter	Value	Parameter	Value
Parallel environments	300	Planning horizon	3
Transitions per variant	1,398,600	Optimization iterations	6
Updates per variant	103,230	Candidate trajectories	512
Batch size	1,024	Elite trajectories	64
Replay capacity (storage rows)	6,000,000	Policy-prior trajectories	24
Learning rate	3×10^{-4}	Minimum sampling std.	0.05
Observation-encoder LR multiplier	0.3	Maximum sampling std.	2
Update-to-data ratio	0.075	Temperature	0.5
Gradient-norm clipping	20	Control frequency	15 Hz
Target-Q Polyak coefficient	0.01	Terminal Q aggregation	Two-critic mean
Discount factor	0.95	Proposal initialization	Prior mean/std.
		Final exploration noise	Off at evaluation

M1–M3: history and gradient-free burn-in are each 4 steps; M0: neither is used. The training rollout is 3 transitions for all variants. Replay capacity is $50 \times 120{,}000$ storage rows, not occupancy.

The first 22,200 transitions are collected randomly without updates; subsequent updates follow the accumulated update-to-data ratio. The learning-rate multiplier of 0.3 applies only to the observation/latent-state encoder. Task, history and contact encoders use the base rate of 3×10^{-4}.

D Evaluation Metrics

D.1 Simulation Success

Let $h_{\mathrm{target},i}$ be the runtime target insertion depth and $u_{z,i,t} = [p_{i,t}^{\mathrm{rel}}]_z$ the held part's axial coordinate in the fixed-part mating frame. For both task types, insertion decreases u_z under the convention in Appendix A.3. Insertion depth is $\ell_{i,t} = \max(0, h_{\mathrm{target},i} - u_{z,i,t})$, with normalized depth $\rho_{i,t} = \ell_{i,t}/h_{\mathrm{target},i}$. The lateral distance is $e_{xy,i,t} = \|[p_{i,t}^{\mathrm{rel}}]_{xy}\|_2$. Let T_i denote the final control-sample index and $\varepsilon_i = \mathrm{clip}(2c_i, 1\,\mathrm{mm}, 3\,\mathrm{mm})$ the lateral tolerance. The success flag checks the minimum depth fraction and maximum lateral error over the final three samples:

$$S_i^{\mathrm{sim}} = \begin{cases} 1, & \min\limits_{j\in\{0,1,2\}} \rho_{i,T_i-j} \geq 0.85 \quad \text{and} \quad \max\limits_{j\in\{0,1,2\}} e_{xy,i,T_i-j} \leq \varepsilon_i, \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

No upper insertion-depth bound is imposed, and episodes continue after first geometric success. The three samples span $2/15 \simeq 0.133$ s at 15 Hz. This operational criterion does not guarantee physical fit or absence of interpenetration. SR is the percentage of successful trials, task-macro-averaged in the main simulation comparison. Hardware uses the 1 s stable-insertion criterion in Appendix F.1.

Relation to the AutoMate criterion. The legacy AutoMate check considered here combines engagement height with a fixed mean keypoint-distance threshold of 20 mm and latches success once a trajectory enters the accepted region. Subsequent withdrawal does not revoke that flag. The criterion above instead checks terminal depth and lateral alignment over consecutive final samples.

D.2 Final Insertion Progress

Final insertion progress (FP) is mean normalized terminal axial depth over all evaluated trials, including successes, failures, and protective stops:

$$\mathrm{FP} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{clip}(\rho_{i,T_i}, 0, 1), \qquad N > 0. \tag{25}$$

Each trial is normalized and clipped before averaging; stopped trials use their last recorded depth. FP measures terminal depth rather than maximum depth reached. It complements SR but does not identify failure causes or establish stable assembly. An empty evaluation set is reported as N/A.

D.3 Contact Diagnostics

In simulation, a near-jam event occurs when raw plug–socket contact-force magnitude is at least 15 N at each of four consecutive control samples, while net insertion-depth increase from the first to last sample is below 0.1 mm. The samples span $3/15 = 0.20$ s at 15 Hz. Progress is signed, so retreat can satisfy this condition; no median filtering or completion masking is applied. Let $J_i = 1$ if any such event occurs during trial i, and $J_i = 0$ otherwise. JR is the percentage of trials with $J_i = 1$. It measures event incidence: a trial can recover and succeed after an event. Near-jam events are distinct from protective stops and do not constitute independently verified mechanical-jam failures.

For recorded three-dimensional force $F_{i,t}$, the conditional statistic is

$$\overline{F}_{\mathrm{peak},J=0} = \frac{\sum_i (1 - J_i) \max_t \|F_{i,t}\|_2}{\sum_i (1 - J_i)}. \tag{26}$$

It excludes torque and trials with near-jam events, and is undefined when no $J = 0$ trials remain. It describes each method's event-free subset, not force exposure over all trials. Protective stops count as failures.

E Additional Simulation Results

E.1 Aggregate and Per-Task Results

Evaluation protocol. For the ten-task study, each method is evaluated separately on the training-size set and the complementary held-out test-size set defined in Appendix A.1. Each assembly task has 100 evaluation trials on each split. Tables 1 and 7, and Figure 7, report only held-out test-set statistics; training-set evaluation results are excluded from these reported comparisons.

The TD-MPC2 component definitions and main results are given in Table 1. M0-to-M1 jointly changes wrench access, history and recurrence, while M2-to-M3 enables two auxiliary losses together. These comparisons do not isolate the effect of each history channel or each auxiliary head.

Complete reported outcomes. Table 7 collects the matched simulation results and the separate policy-transfer comparison, including JR omitted from the main-text table. Metric definitions are given in Appendix D.

Table 7: Held-out test-set simulation outcomes accompanying Section 4.2. The upper block contains matched baselines and component ablations; the lower block is the separate SRSA-style policy-transfer comparison against FACA. FP is final insertion progress over all evaluated trials, averaged after normalization by target depth and clipping to $[0, 1]$; JR is near-jam event incidence. $\overline{F}_{\text{peak},J=0}$ is mean episode-peak force over trials without a detected near-jam event.

Method	SR (%)↑	FP↑	JR (%)↓	$\overline{F}_{\text{peak},J=0}$ (N)↓
AutoMate-style PPO	54.1	0.63	30.6	20.9
Task-PPO	53.6	0.65	34.1	20.1
Task-TD-MPC2 (M0)	66.0	0.73	25.2	19.6
+ recurrent history (M1)	69.0	0.77	23.1	16.4
+ contact encoder (M2)	71.8	0.79	22.4	13.5
FACA (M3)	**76.1**	0.84	**20.4**	**12.9**
Separate policy-transfer comparison against FACA				
SRSA-style policy transfer	68.3	0.69	30.1	19.2

AutoMate-style PPO omits explicit task conditioning; Task-PPO receives the descriptor. All numerical comparisons use evaluation in our environment.

Per-task results. Figure 7 shows overall and per-task success rates for the policy-level comparison.

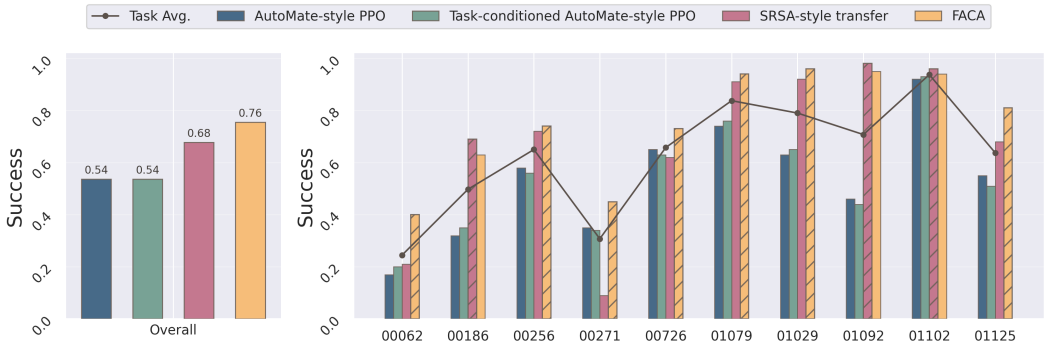


Figure 7: Overall and per-task success rates on held-out size conditions of the ten axial assembly tasks. The left panel shows overall rates, and the right panel shows rates by task ID. The Task Avg. curve averages the four methods within each task.

E.2 Qualitative Contact-Recovery Example

Figure 8 compares M0 and FACA (M3) on the hole-onto-peg task 00186 from matched recorded initial poses. Following contact and adjustment, FACA advances insertion while M0 remains stalled in the displayed interval. This example compares the complete configurations.

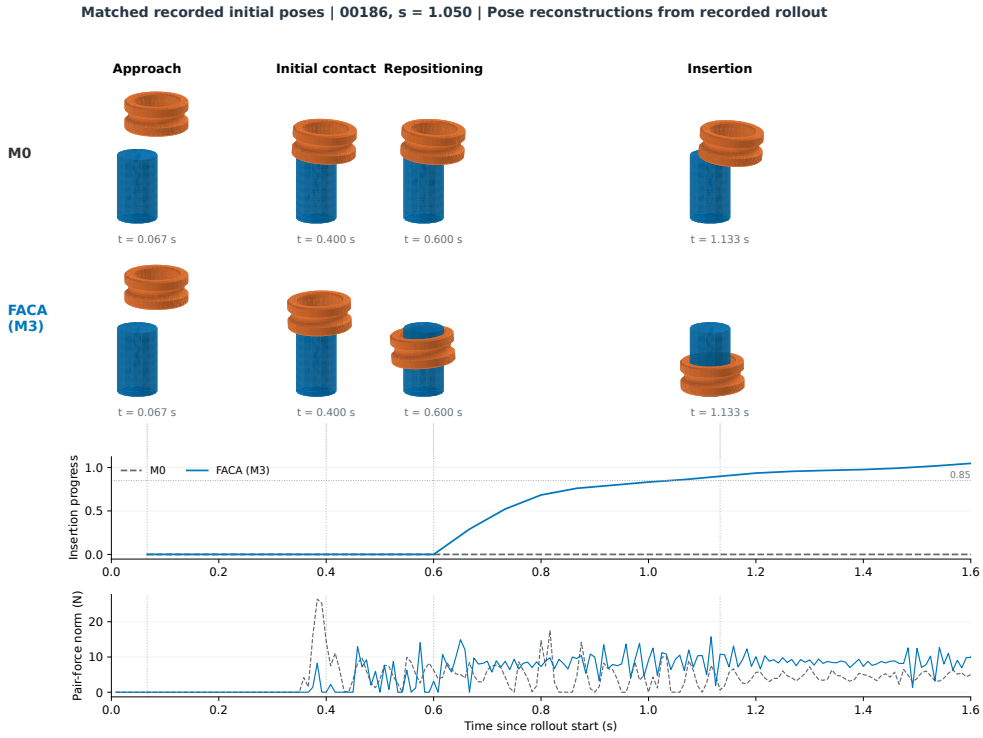


Figure 8: Contact recovery for assembly 00186 at $s = 1.050$, from separately recorded rollouts with matched initial poses (0–1.6 s excerpt). Top: reconstructed poses of the orange held hole part and blue fixed peg, linked by dotted lines to the selected times. Middle: insertion depth divided by target depth; 0.85 marks only the depth threshold. Bottom: plug–socket pair-force norm, distinct from the wrist-wrench input. The excerpt illustrates contact recovery; terminal success is assessed using the full evaluation protocol.

E.3 Task-Descriptor Study

This independent comparison uses the same force-free backbone and target-relative state inputs on 00186 (hole-onto-peg) and 00271 (peg-in-hole). Each family uses five scales per split: training $s \in \{0.90, 0.95, 1.00, 1.05, 1.10\}$; unseen interpolation $s \in \{0.925, 0.975, 1.025, 1.075, 1.085\}$; and extrapolation $s \in \{0.85, 0.875, 1.125, 1.15, 1.175\}$. There are ten conditions per split across the two families.

Table 8: Descriptor comparison on two assembly families. The first row omits only the descriptor branch; the full descriptor adds only mating type to scale, axial extent, and clearance. Seen denotes training scales. Held-out average is the mean interpolation and extrapolation SR.

Task encoding	Seen sizes SR (%)↑	Unseen interpolation SR (%)↑	Unseen extrapolation SR (%)↑	Held-out avg. SR (%)↑
No explicit task-descriptor	55.3	50.2	28.4	39.3
Task type only	56.4	50.1	26.1	38.1
Scale + axial extent + clearance	62.1	54.3	48.7	51.5
Full task descriptor p	**65.1**	**56.2**	**52.4**	**54.3**

F Real-Robot Protocol and Calibration

F.1 Hardware, Collection and Calibration

The setup uses a Franka Research 3 with a wrist force/torque sensor, Cartesian translational actions and controller-maintained orientation. The operator places the grasped part at the assembly start position and records its TCP position p_{ref}. Reference error ϵ_i is sampled once per trial and held fixed: $\tilde{p}_{\text{ref},i} = p_{\text{ref}} + \epsilon_i$. Runtime relative position is $\Delta p_{i,t} = p_{\text{TCP},i,t} - \tilde{p}_{\text{ref},i}$, in a common coordinate basis. Reference error follows the simulation distribution and range, separately from initial TCP randomization. The reference is a taught TCP position, not a measured socket pose. Each method has 50 evaluation trials over ten tasks and ten sessions. Reported offsets reach 5% of peg radius laterally and 3° angularly, with a reported 3 mm frame-calibration error. Hardware success requires completed insertion to remain stable for at least 1 s continuously. Protective stops count as failures; no policy recovery follows a stop, and progress and force records end at termination.

Full-FT updates only the encoder and base dynamics; the reward, value, policy prior and auxiliary heads remain fixed. Full-FT and Auto-adapt use the same ten autonomous collection episodes from a single anchor task: 740 transitions in total, 2,000 updates, and the same replay sampling seed. HIL-adapt collects ten anchor-task episodes with the same transition budget and up to 111 operator actions replacing autonomous proposals. The correction trigger is at least 10 N tared TCP force and less than 0.1 mm progress over four control samples. Replay stores executed actions and resulting responses; collection episodes are excluded from frozen evaluation. Equal transition budgets do not imply equal collection time or operator effort. Each adapter is shared across all ten evaluation tasks without per-task calibration.

For Auto/HIL calibration, only the residual adapter is updated; base encoders, dynamics, reward, Q, auxiliary heads, and policy prior remain fixed. The adapter has 389,129 parameters (19.0% of the full trainable model). The effective residual and adapted transition are

$$\begin{aligned} A_\omega^{\text{eff}}(v) &= g\alpha_{\text{eff}}\,\text{clip}(\Delta z_\omega(v), -\delta, \delta), \\ \hat{z}_{t+h+1|t} &= D_\phi(\hat{z}_{t+h|t}, a_{t+h|t}, c_{\text{task}}) + A_\omega^{\text{eff}}(v_{t+h|t}). \end{aligned} \tag{27}$$

Here $\hat{z}_{t|t} = z_t$, $h = 0, \ldots, H_{\text{mpc}} - 1$, and g, α_{eff}, and δ denote the gate, residual scale, and componentwise clipping bound. Inputs v use no future measured wrench. The residual is added without post-add SimNorm, and the corrected latent is propagated at every imagined step, including policy-prior candidate generation.

For Auto/HIL calibration, the latent-consistency component uses Eq. (20) with adapted dynamics, a frozen encoder, and the same zero initialization and burn-in. Correction transitions introduce no separate imitation objective.

F.2 Evaluation and Termination Outcomes

Table 2 reports real-robot outcomes. Both the near-jam indicator J_i and conditional peak force use the tared wrist-force signal. Table 9 gives termination counts. Other failures are 50 minus successes and protective stops, without individual failure-mechanism attribution. Near-jam diagnostics can co-occur with protective stops.

Table 9: Per-method termination outcomes. Other failures are residual counts from the reported success and protective-stop totals.

Method	Success	Protective-stop failure	Other failure	Total
Sim-only	26	18	6	50
Full-FT	30	15	5	50
Auto-adapt	34	12	4	50
HIL-adapt	41	5	4	50
Total	131	50	19	200